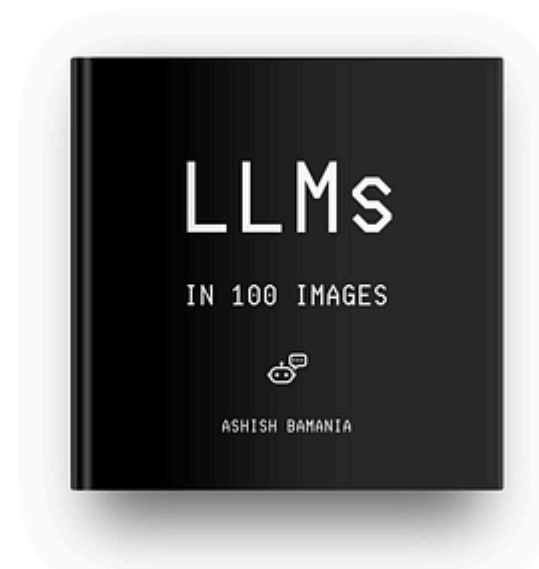
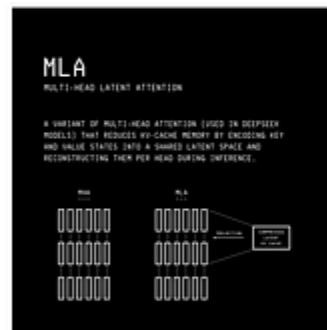


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All The Basics That You Need To Know About LLMs

We're living in an amazing time where understanding AI can open so many doors. Learning about LLMs might be your smartest investment today.

Dr. Ashish Bamanian

a11y-light · August 24, 2025 (Updated: August 24, 2025) · Free: No

LLMs are one of the most successful AI technologies to date.

The LLM market is expected to reach a total value of \$82 billion by 2033.

As of 2025, 67% of organizations worldwide have adopted LLMs to support their operations with generative AI.

If you're new to LLMs and would love to deepen your knowledge, this article will help you do that.

Let's begin!

What Are LLMs?

Large Language Models (LLMs) are AI systems that are trained on vast amounts of text data to understand and generate human-like language.

During training, they learn patterns, relationships, and structures in language by analyzing billions of text examples from books, articles, websites, and other

This gives them an understanding of grammar and semantics in human language.

Some of the popular LLMs used today are:

- GPT-4o from OpenAI (in the form of ChatGPT)
- Claude Sonnet 4 from Anthropic
- Gemini 2.5 Flash from Google

These models are proprietary, which means that their internal details (weights, parameters, training data, training methods) aren't publicly available.

The most widely used open-weight models, where model weights are publicly available, are:

- Llama by Meta
- DeepSeek-V3 by DeepSeek
- Mistral Medium 3 by Mistral AI

LLM

LARGE LANGUAGE MODEL

AN AI SYSTEM TRAINED ON VAST AMOUNTS OF TEXT DATA TO UNDERSTAND AND GENERATE HUMAN-LIKE LANGUAGE FOR TASKS SUCH AS CONVERSATION, WRITING, AND REASONING.



CHATGPT



GEMINI



LLAMA



DEEPSEEK

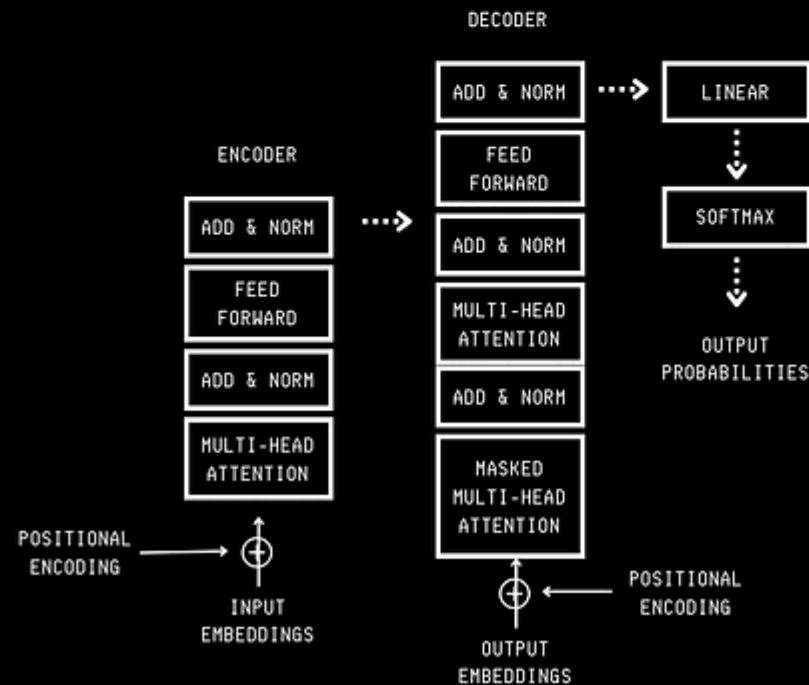
What Powers An LLM?

The Transformer architecture is the backbone of all popular LLMs that we use today.

Transformer was developed through research by Google in 2017.

TRANSFORMER

A NEURAL NETWORK ARCHITECTURE DESIGNED FOR HANDLING SEQUENTIAL DATA, RELYING ON SELF-ATTENTION MECHANISMS TO CAPTURE DEPENDENCIES BETWEEN INPUT ELEMENTS WITHOUT REGARD TO THEIR DISTANCE IN THE SEQUENCE.



one after another (sequentially).

This is achieved through its mechanism called **Self-attention**, which helps figure out how each word relates to every other word in the text sequence.

SELF-ATTENTION

SCALED DOT-PRODUCT ATTENTION

A MECHANISM THAT LETS AN LLM WEIGH AND AGGREGATE INFORMATION FROM ALL TOKENS IN A SEQUENCE (INCLUDING ITSELF) TO COMPUTE A NEW REPRESENTATION FOR EACH TOKEN, ENABLING IT TO CAPTURE CONTEXT, DEPENDENCIES, AND RELATIONSHIPS ACROSS THE SEQUENCE.

$$\text{ATTENTION}(Q, K, V) = \text{SOFTMAX} \left(\frac{Q * K^T}{\text{SQRT}(D)} \right) * V$$

WHERE:

- Q -> QUERIES
- K -> KEYS
- V -> VALUES
- D -> DIMENSION OF KEYS

What Is GPT?

GPT, or **Generative Pre-trained Transformer**, is one of the earliest and most widely known LLMs.

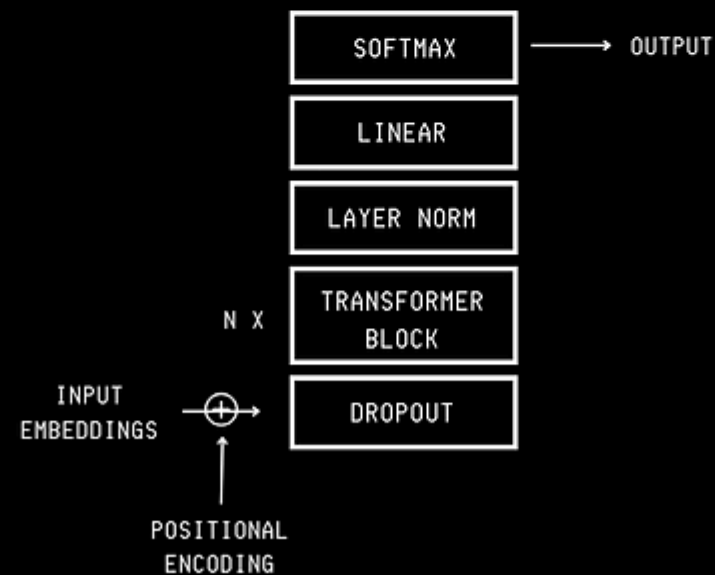
GPT was born out of research from OpenAI in 2018, just one year after Google introduced the Transformer architecture.

Its successor, ChatGPT, is one of the most popular LLMs used today.

GPT

GENERATIVE PRE-TRAINED TRANSFORMER

A LANGUAGE MODEL THAT USES THE TRANSFORMER ARCHITECTURE TO GENERATE HUMAN-LIKE TEXT BY PREDICTING THE NEXT WORD IN A SEQUENCE, PRE-TRAINED ON VAST TEXT DATA AND FINE-TUNED FOR SPECIFIC TASKS.



This process is called **Autoregression**, which means that each word is generated based on the previous ones.

AUTOREGRESSION

NEXT-TOKEN PREDICTION

A MODELING APPROACH WHERE EACH OUTPUT IS PREDICTED BASED ON PREVIOUS OUTPUTS, COMMONLY USED IN LANGUAGE MODELS TO GENERATE THE NEXT TOKEN FROM PRIOR TOKENS.



This seems strange because it should have been a word/ sentence that goes into the GPT for it to produce the next word.

The truth is that LLMs do not understand English (or any other human language).

Any word/ sentence in English has to be first broken down into small pieces called **Tokens** in a process called **Tokenization**.

In LLMs like ChatGPT, this is done using a Tokenization algorithm called **Byte Pair Encoding**.

BPE

BYTE-PAIR ENCODING

A TOKENIZATION TECHNIQUE THAT ITERATIVELY MERGES THE MOST FREQUENTLY OCCURRING PAIRS OF CHARACTERS OR TOKENS TO BUILD A VOCABULARY OF SUBWORD UNITS. OPENAI'S 'TIKTOKEN' LIBRARY IMPLEMENTS A FAST, MODEL-OPTIMIZED VERSION OF BPE FOR TOKENIZING TEXT.

"NEWEST" $\xrightarrow{\text{TIKTOKEN}}$ ["NEW", "EST"]

"HAPPINESS" $\xrightarrow{\text{TIKTOKEN}}$ ["H", "APPINESS"]

Embeddings are high-dimensional vector representations that capture the semantic meaning and relationships between different words/ sentences.

Words with similar meanings have embeddings that are closer to each other in a higher-dimensional space.

This is shown below, where the embedding of "*Apple*" is closer to that of "*Orange*" than "*Pen*".

EMBEDDINGS

DENSE VECTOR REPRESENTATIONS OF TOKENS, WORDS, OR SENTENCES THAT CAPTURE THEIR SEMANTIC MEANING AND RELATIONSHIPS IN A CONTINUOUS VECTOR SPACE.



This could lead to issues because in a language like English, the positioning of words is important for conveying meaning.

This is why **Positional Encodings** are used to combine the positional information of different words/ tokens in a sentence with the input embeddings of those words/ tokens.

POSITIONAL ENCODING

A TECHNIQUE USED IN TRANSFORMERS TO INJECT INFORMATION ABOUT THE ORDER OF TOKENS INTO THE MODEL, ENABLING IT TO CAPTURE SEQUENCE STRUCTURE DESPITE HAVING NO INHERENT RECURRENCE.

TOKEN	POSITION	TOKEN EMBEDDING	POSITIONAL ENCODING	FINAL INPUT TO TRANSFORMER
I	0	[0.2, 0.5]	[0.1, 0.3]	[0.3, 0.8]
LOVE	1	[0.6, 0.1]	[0.4, 0.6]	[1.0, 0.7]
AI	2	[0.9, 0.3]	[0.7, 0.9]	[1.6, 1.2]

Training An LLM To Generate Text

The first step to go from zero to a text-generating LLM is **Pretraining**.

During this stage, the LLM learns by processing massive unlabelled text datasets.

At each step, it is given the context (i.e., the previous words/ tokens) and asked to predict the following word/ token.

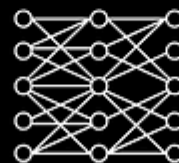
This makes it gradually learn grammar, facts, and common-sense reasoning.

PRETRAINING

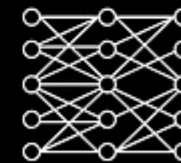
THE INITIAL PHASE WHERE AN LLM LEARNS GENERAL LANGUAGE PATTERNS AND KNOWLEDGE BY TRAINING ON VAST AMOUNTS OF UNLABELLED TEXT DATA BEFORE BEING FINE-TUNED FOR SPECIFIC TASKS.



UNLABELLED
TEXT DATA



UNTRAINED LLM



PRE-TRAINED LLM

These tasks could range from the model answering questions, summarizing documents, or following instructions more reliably.

This step is called **Supervised Fine-tuning (SFT)**.

SFT

SUPERVISED FINE-TUNING

THE PROCESS OF TRAINING A PRE-TRAINED LANGUAGE MODEL ON LABELLED INPUT-OUTPUT PAIRS TO SPECIALIZE IT FOR SPECIFIC TASKS OR BEHAVIOURS.



PRETRAINED LLM



- NAMED ENTITY RECOGNITION (NER)
- QUESTION ANSWERING (QA)
- DOMAIN ADAPTATION
- TEXT CLASSIFICATION
- INSTRUCTION FOLLOWING
- SAFETY AND ALIGNMENT
- TASK-SPECIFIC SKILLS (CODE GENERATION)

As an example, if you ask an LLM, "*When is Christmas?*", it might respond with "*Isn't it 25th December?*".

Although this response is correct, you'd prefer something which sounds more polite, like "*Christmas is celebrated on 25th December every year.*"

This is made possible using a technique called **Reinforcement Learning from Human Feedback (RLHF)**.

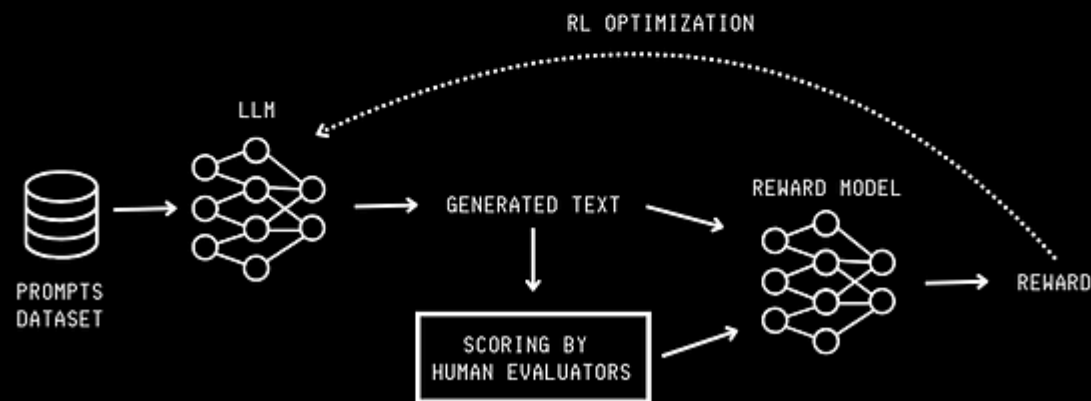
RLHF aligns LLMs with human values, preferences, and expectations by using datasets of human judgments that guide them on what responses are considered better.

It is the key technique that enables modern LLMs, such as ChatGPT, to achieve high conversational quality and safety.

RLHF

REINFORCEMENT LEARNING FROM HUMAN FEEDBACK

A FINE-TUNING TECHNIQUE WHERE AN LLM IS TRAINED TO ALIGN WITH HUMAN PREFERENCES BY OPTIMIZING ITS OUTPUTS BASED ON FEEDBACK-RANKED RESPONSES USING A REWARD MODEL AND REINFORCEMENT LEARNING.



How Do You Get Better Responses From An LLM?

Prompting is a popular technique that can help you get responses from an LLM, and a whole field called **Prompt Engineering** has emerged around this practice.

PROMPT ENGINEERING

THE PRACTICE OF DESIGNING AND OPTIMIZING INPUT PROMPTS TO GUIDE AN LLM'S BEHAVIOR AND IMPROVE THE RELEVANCE, ACCURACY, OR CREATIVITY OF ITS OUTPUTS.

WITHOUT PROMPT ENGINEERING



"EXPLAIN RELATIVITY."



TECHNICAL OR DRY OUTPUT

WITH PROMPT ENGINEERING



"EXPLAIN EINSTEIN'S THEORY OF RELATIVITY TO A 12-YEAR-OLD USING SIMPLE METAPHORS."



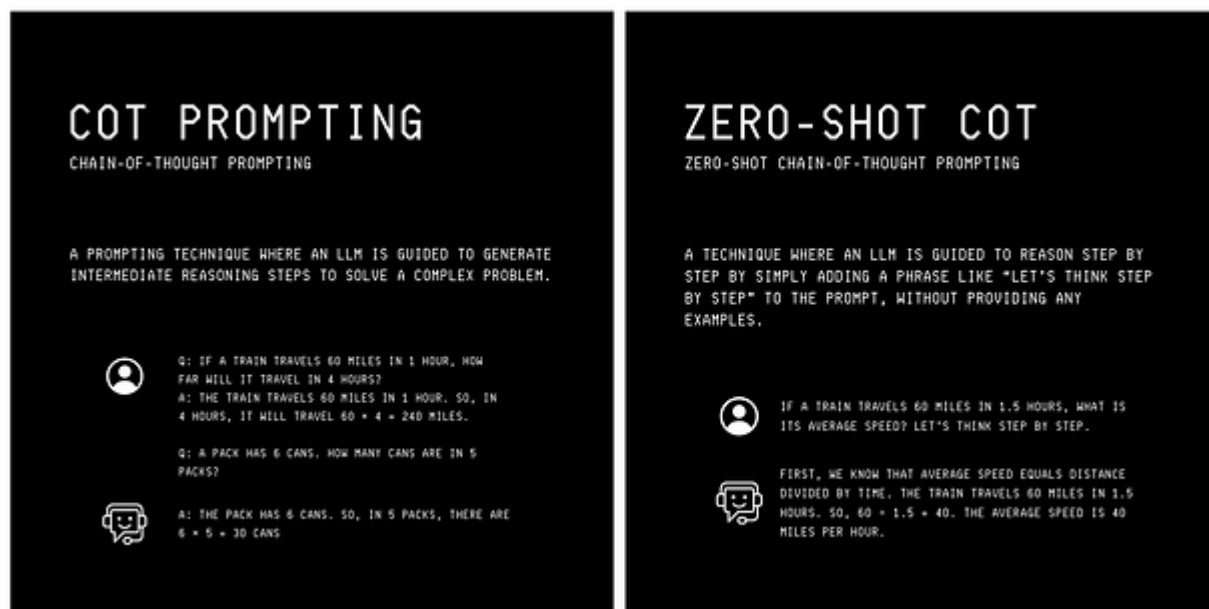
FRIENDLY, AGE-APPROPRIATE EXPLANATION

- **Zero-shot Prompting:** where one directly instructs an LLM to perform a task
- **Few-shot Prompting:** where one provides a few examples related to the task to be achieved, along with the instructions to complete the task. This usually results in better responses from an LLM.



Alongside these, many specialized techniques for prompting have been introduced, and one of them is called **Chain-of-Thought (CoT) Prompting**.

When following Chain-of-thought (CoT) prompting, the LLM is instructed to reason step by step before providing an answer.



There is also a way of further training LLMs so that they internalize this Chain-of-thought approach. This helps them *think* and reason better when responding to complex problems.

This is achieved by training LLMs on massive datasets of examples of prompts and their Chain-of-thought responses, using reinforcement learning.

The resulting LLMs are referred to as **Large Reasoning Models (LRMs)**. These models take their time thinking before answering a query.

Freedium

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-
- GPT-3 and GPT-4 by OpenAI
 - Claude Opus 4 by Anthropic
 - DeepSeek-R1 by DeepSeek

LRM

LARGE REASONING MODEL

AN ADVANCED LANGUAGE MODEL DESIGNED NOT JUST FOR LANGUAGE UNDERSTANDING, BUT FOR PERFORMING COMPLEX, MULTI-STEP REASONING TASKS BY LEVERAGING MEMORY, PLANNING, AND STRUCTURED THINKING CAPABILITIES.



OPENAI 01



GEMINI 2.0
FLASH THINKING



DEEPSEEK-R1

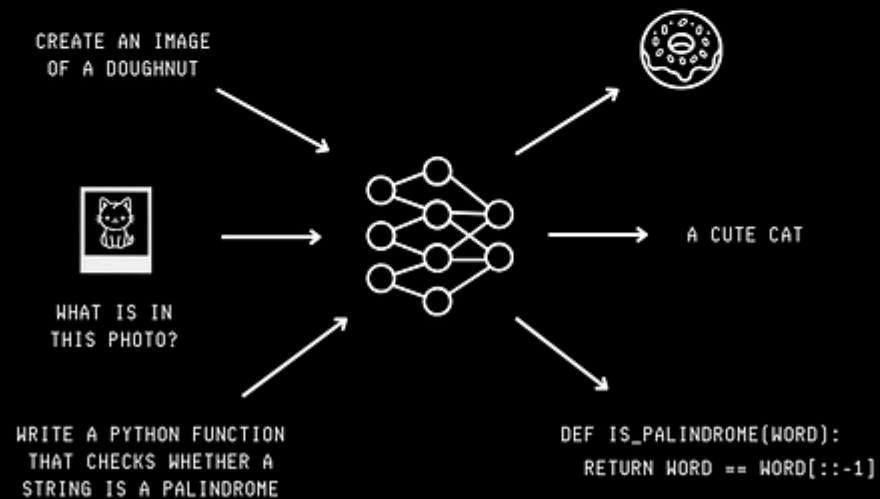
Modern-day LLMs Are Multi-Modal

LLMs are not only text-generators, but they can do much more than that.

Modern-day LLMs are multi-modal. This means that they can work with data from different modalities (audio, images, and videos) as their inputs and outputs.

MULTIMODAL LLMs

LLMs THAT CAN UNDERSTAND AND GENERATE CONTENT ACROSS MULTIPLE DATA TYPES (TEXT, IMAGES, AUDIO, VIDEO) FOR BOTH INPUT AND OUTPUT, ENABLING RICHER HUMAN-AI INTERACTION BEYOND TEXT-ONLY COMMUNICATION.



What Is Agentic AI?

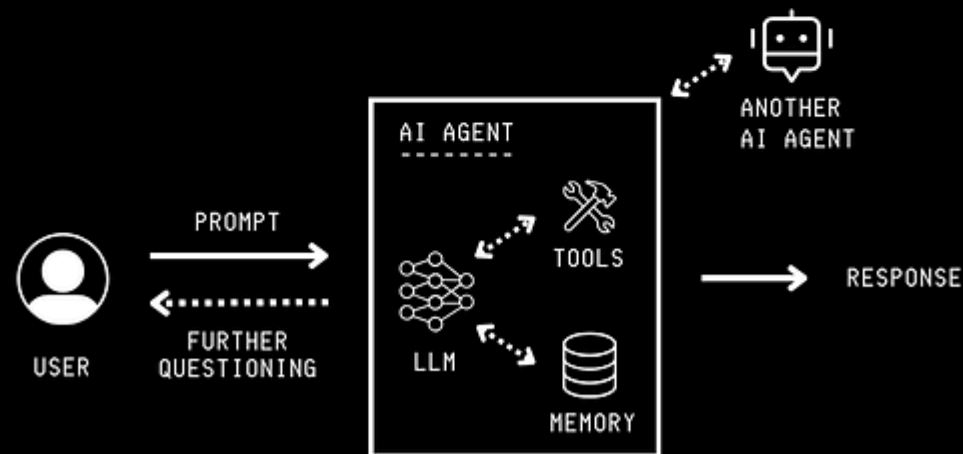
Modern-day LLMs can also have agency and be autonomous. This gives rise to their use as **AI agents**.

Being *Agentic* means that LLMs can act as the brain of systems, where, when given a task, they can:

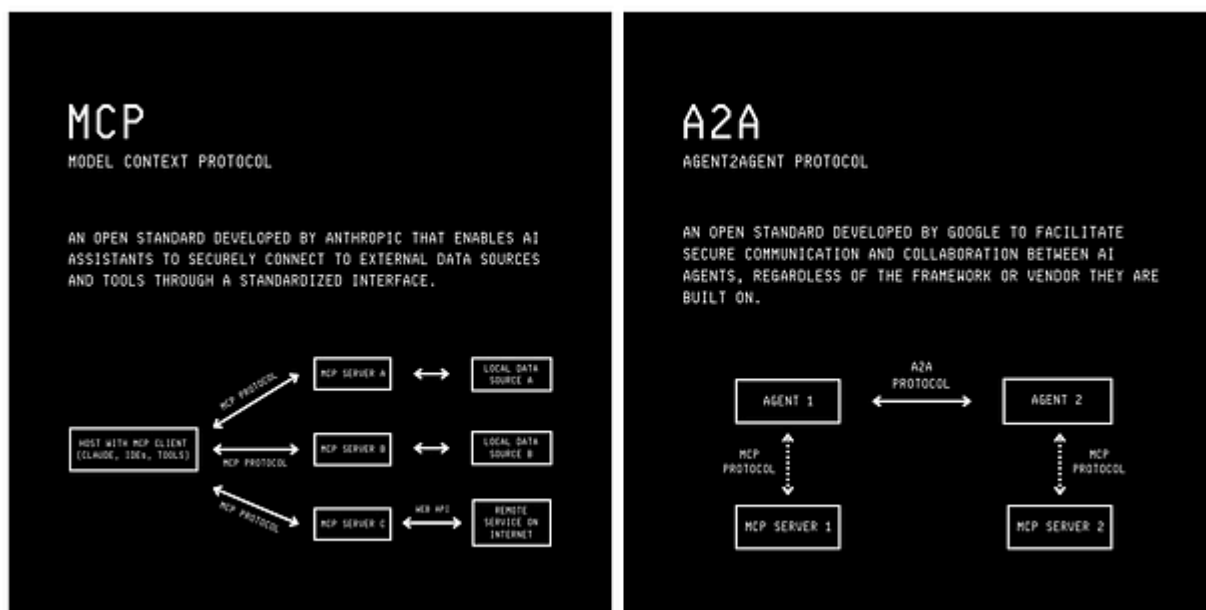
- Reason and plan the approach to complete the task
- Use task-specific tools to interact with environments and other agents
- Get feedback and amend their approach for completing the task

AI AGENTS

AUTONOMOUS SYSTEMS POWERED BY LANGUAGE MODELS THAT CAN REASON, PLAN, USE TOOLS, INTERACT WITH ENVIRONMENTS, ACCESS MEMORY, AND REFINE THEIR ACTIONS THROUGH FEEDBACK TO ACHIEVE USER-DEFINED GOALS.



1. **Model Context Protocol (MCP)**: This protocol, developed and open-sourced by Anthropic, enables agents to seamlessly access and work with external data sources, APIs, tools, and applications.
2. **Agent2Agent (A2A) Protocol**: This protocol, developed and open-sourced by Google, enables multiple independent AI agents to collaborate and work towards a given task.

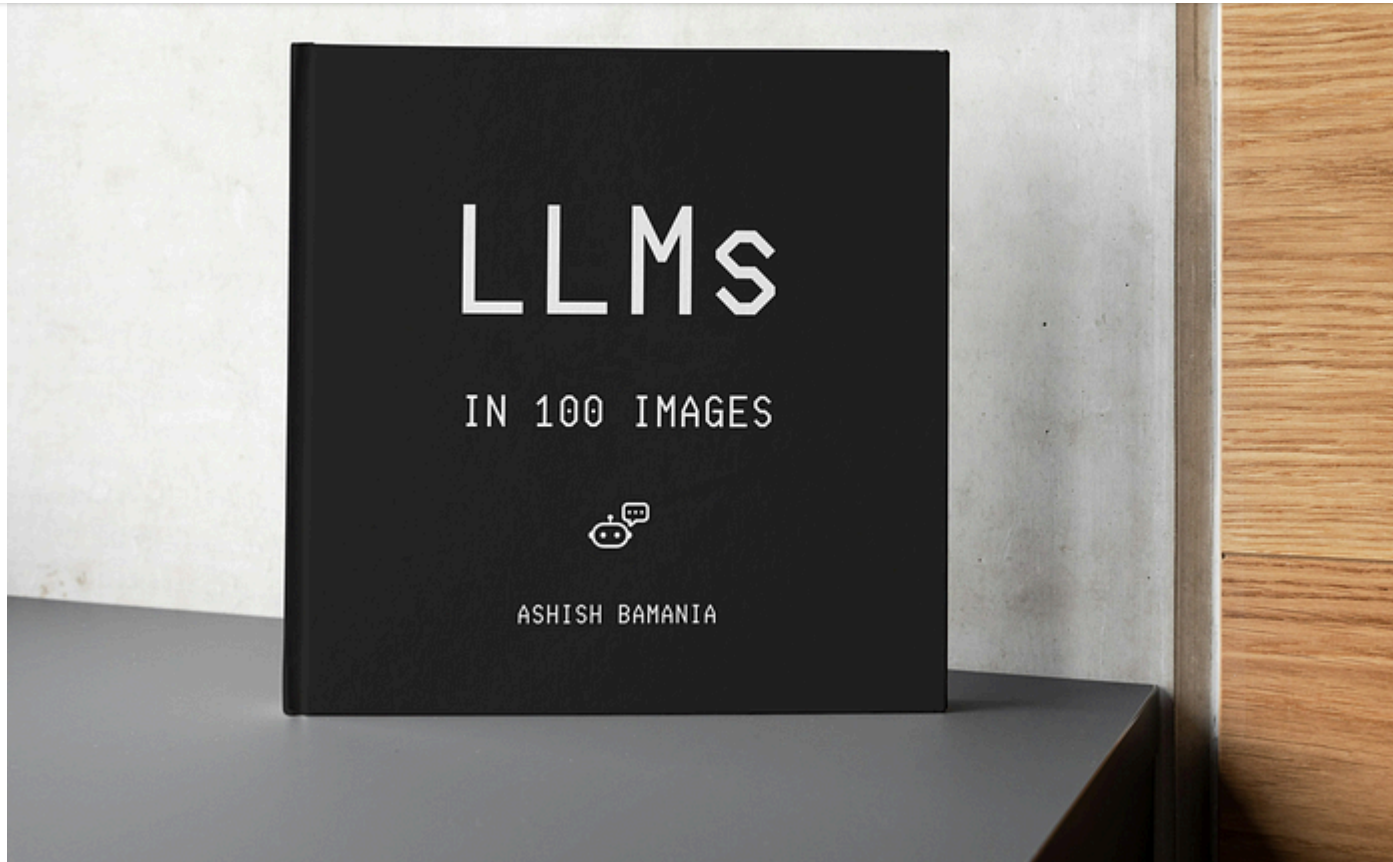


This article was originally published in 'Engineering Leadership' by Gregor Ojstersek.

All the visuals used in this article are obtained from my book called "LLMs In 100 Images".

It is a collection of 100 easy-to-follow visuals that describe the most important concepts you need to master LLMs today.

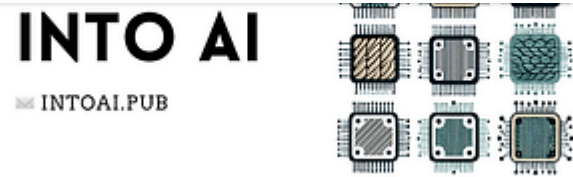
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